



SmartVista Instant Payment System

Guide – CAS API Services

**PRODUCT
DOCUMENTATION**

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bpcbt.com

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Version

Version	Release Type	Release Date	Summary Of Changes	Impact Analysis
1.0.0	Initial Release	10-06-2024	Initial Release	Initial draft
1.0.1	Minor change	02-07-2024	Fields added to DB schema and their logics in entity tables	

Glossary

Term	Definition
CAS	Customer addressing schema
UUID	Universally Unique Identifier
IPS	Instant payment system
OTP	One time password
Customer	Company or individual, who opened an account with bank
Alias	an alias is an easily recognizable identifier, such as an email address or phone number
FA	Financial address
MSISDN	Mobile Station International Subscriber Directory Number is the unique phone number assigned to a mobile subscriber in a cellular network
Open API editor	An API editor is a tool that allows you to create, edit, and manage API definitions, typically in formats like OpenAPI or Swagger

1 Introduction

CAS enables registered customers to initiate payments using aliases of registered customers instead of account numbers. An alias serves as an easy-to-remember identifier, such as a national ID, mobile phone number, driver's license, or similar identifier. Each alias is linked within the CAS to the customer's registration record and to a list of eligible accounts associated with the customer.

CAS operates in a centralized mode, maintaining all necessary account information (IBAN, customer name) centrally within the IPS.

1.1 Document purpose

The main purpose of this document is as follows:

- To describe the CAS data model and flows.
- A guide to APIs involved in CAS use case.
- To outline the description about the logics involved in CAS.

2 CAS Database Schema

2.1 CAS Data Model

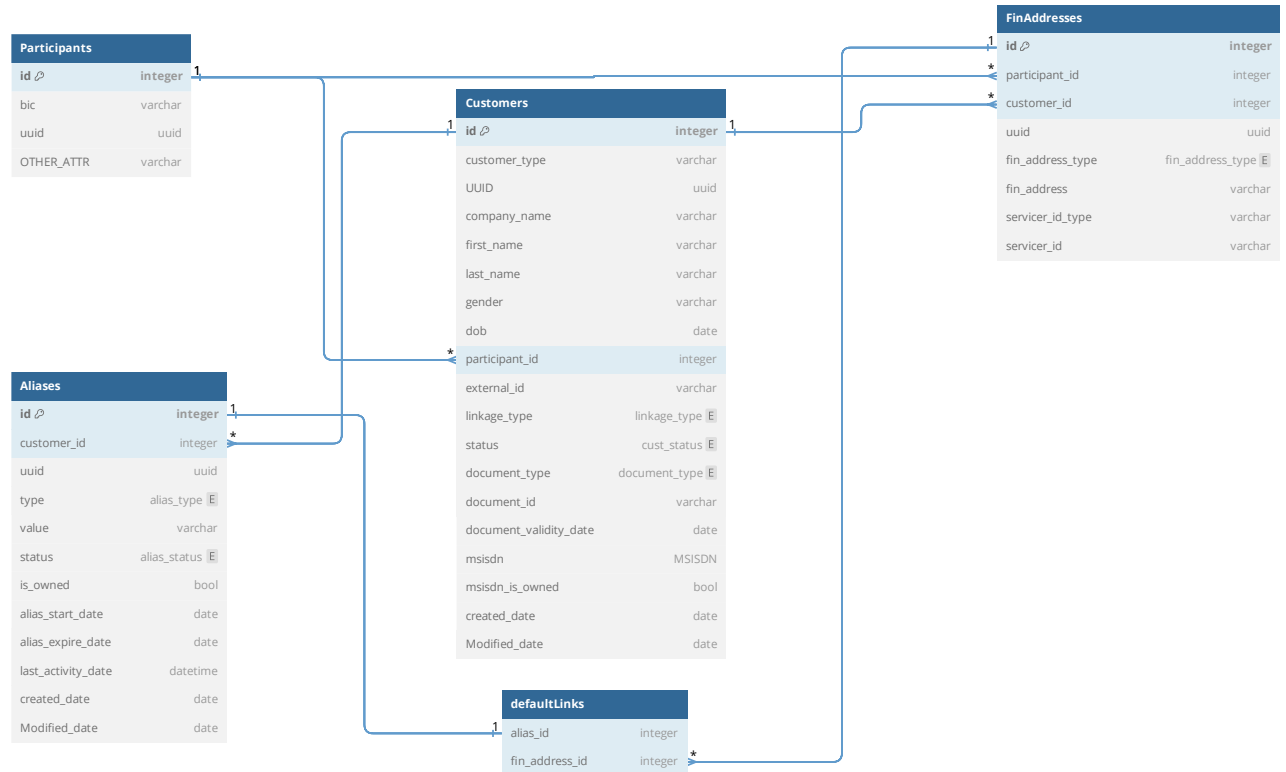


Figure 2-1 cas data model

2.2 Customers Entity

Field Name	Comments
ID	Unique identifier
Customer Type	Defines the type of customer it can be individual or company. If type selected is Individual which is also a default value, then first name and last name is mandatory. If type selected is Company the company name is mandatory
UUID	Unique identifier

First name	Customer First name if type selected individual
Last name	Customer Last name if type selected individual
Company name	Name of company if type selected company
DOB	Customer Date of Birth
Gender	Male/Female/Other
Participant ID	Refer to which participant customer belongs to
External ID	Customer External Identification
Linkage Type	Universal linkage: All aliases owned by customer point to the same default FA. Individual Linkage: Every alias can have its independent default FA from the list of FAs that belong to the same customer.
Status	Status of customer (Active , Inactive , Delete)
Document Type	Type of document (National id, Passport, Driver License)
Document ID	Value of Document Type
Document Validity Date	Expiry date of provided document type
MSISDN	Phone number
MSISDN IS Owned	Verify alias ownership is applied. E.g. OTP via SMS for

	MSISDN ownership. It can be either Yes or NO.
Created Date	Date of creation
Modified Date	Last date of Modification

Table 1 customer entity

2.3 Aliases Entity

Field Name	Comments
ID	Unique Identifier
UUID	Unique Identifier
Customer ID	Reference to customer entity
Status	Status of Alias (Active , Inactive , Deleted)
Alias Type	Email, phone number,Free Format(no validation required).
Alias Value	Alias value
Alias Start Date	Date at which Alias be available for performing transactions. It can be different than created date.
Alias Expire Date	Expiry date of Alias. Each end of month, we shall check for Alias that might be expiring the next month. A notification will be generated and send to the customer/company to notifies that expiry date of his alias is reaching.
MSISDN IS Owned	Verify alias ownership is applied. E.g. OTP via SMS for MSISDN ownership. It can be either Yes or NO.
Last Activity Date	If an alias has not been used (no Alias resolution has been made on the API) for a predefined period, the alias would be set to inactive. Thus, last activity date will

	monitor the last activity on alias. The predefined period of inactivity should be configurable.
Created Date	Date of creation
Modified Date	Last date of Modification

Table 2 alias entity

2.4 FinAddresses Entity

This table keeps the financial address such as account number , wallet id of the customer.

Field Name	Comments
ID	Unique identifier
Participant ID	Reference to Participant entity
Customer ID	Reference to Customer entity
UUID	Unique identifier
Finaddress Type	Account, IBAN, wallet
Finaddress	Value of Account/IBAN/Wallet
Servicer ID Type	BIC
Servicer ID	BIC value

Table 3 finaddress entity

3 API Authorization

In order to authorize requests, IPS API uses Password grant type authentication flow of OAuth 2.0 enhanced with signed JWT token as a second factor authentication.

Every business service request (e.g. verification request, payment request, etc.) must be authenticated via a valid access token provided in HTTP header '*Authorization: Bearer <access_token>*'.

NOTE: In some cases certificate authentication might be absent and authentication will be based upon username/password authentication.

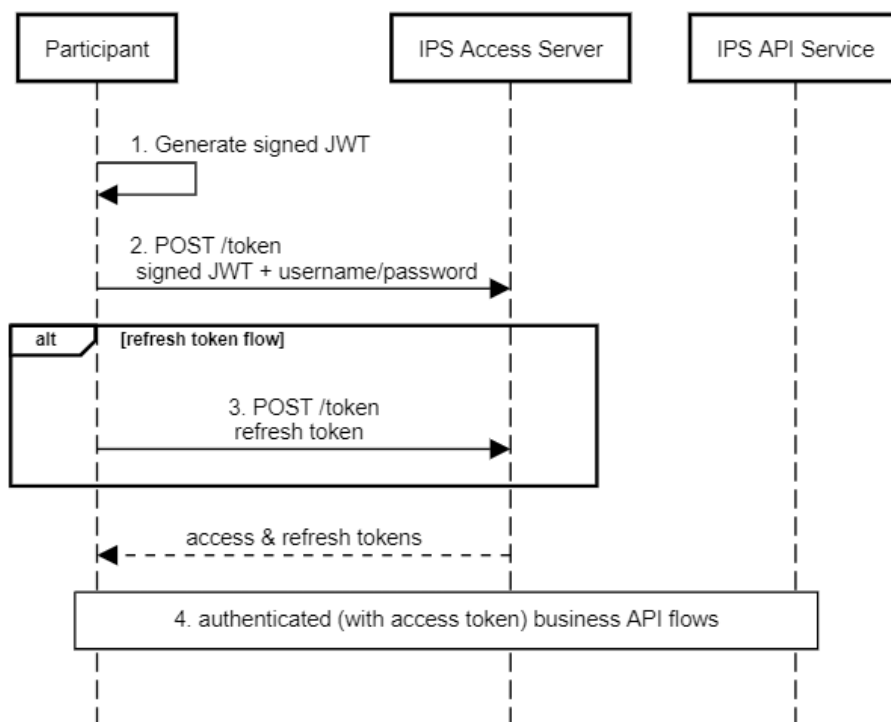


Figure 3-1 api authorization

The IPS API consumer (participant) should undertake the following steps to obtain the valid access token:

1. Generate a client JWT token.
Check Signed JWT token for details.
2. Get an access token and a refresh token from access server
Check Get access token using member credentials and signed JWT for details.
3. Update the access token using the refresh token on a periodic basis as per the parameters provided in the response for the token request

Check Get access token using refresh token for details.

For more details, please refer to **sv_ip__how-to_specification_en_core**.

The member credentials are also used to restrict access to information. Example the initiating participant will have access to only its customers information.

Note: This check will not be applicable for Alias resolution API. As Participant X may request Alias resolution for a customer pertaining to participant Y.

4 Use Cases

4.1 One Step Registration

The one-step registration API is a streamlined process that enables participants to efficiently create a customer profile, an alias, and a financial address in a single request to the CAS service. This approach simplifies the user experience by consolidating multiple actions into one, thereby reducing the complexity and time required for registration.

Implementing a one-to-one link between a phone number and a financial account simplifies the registration process, allowing for seamless integration of services. This approach minimizes the need for customers to manage multiple aliases and accounts, thereby enhancing the user experience. By automating the alias registration through a single-step API, participants can offer a more straightforward and efficient service, focusing on convenience and ease of use.

4.1.1 Use case scheme

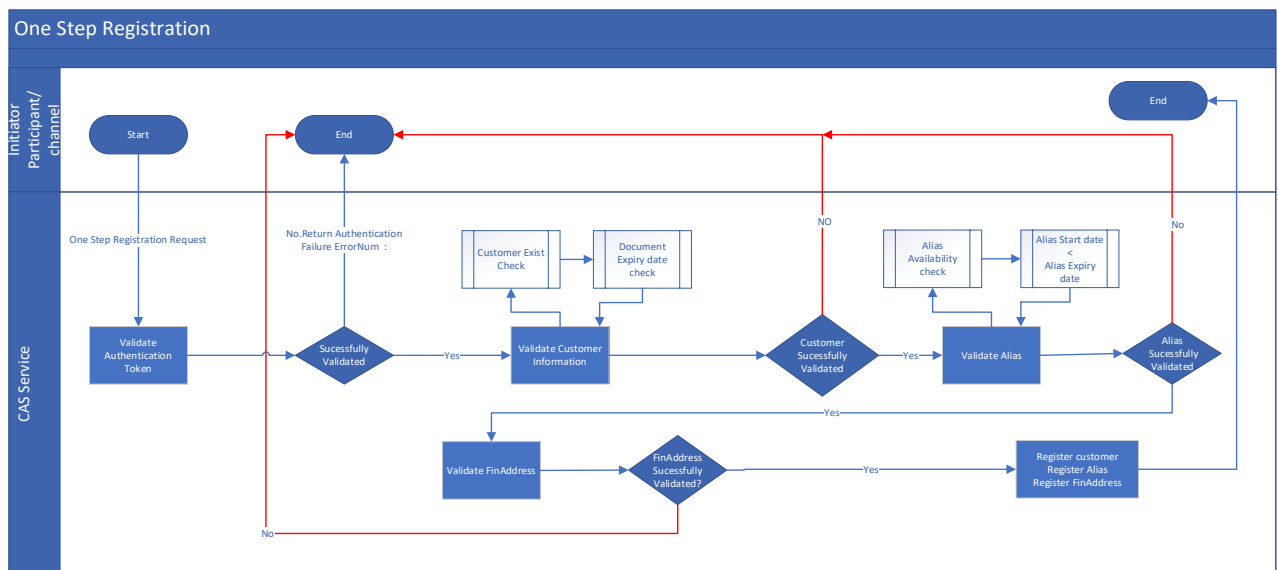


Figure 4-1 one step registration design

4.1.2 API Definition

API Name	createOneStepCustomer
API Version	1.0.0

Endpoint	/cas/v2/customers/one-step
Method	POST

Table 4 one step registration api definition

4.2 Customer Modular Management

In the registration process described, the creation of a customer entity is an independent step, which allows for a modular approach to data management. This separation of processes enhances flexibility, as aliases and financial addresses can be added or updated without affecting the foundational customer record.

Such a system design can improve data integrity and streamline the management of customer information by isolating each component of the registration process.

4.2.1 Customer Entity Management

4.2.1.1 Create Customer

Use to create the customer entity. In case of successful response, CAS service return the customer object including the system generated Customer UUID

4.2.1.1.1 Use case scheme

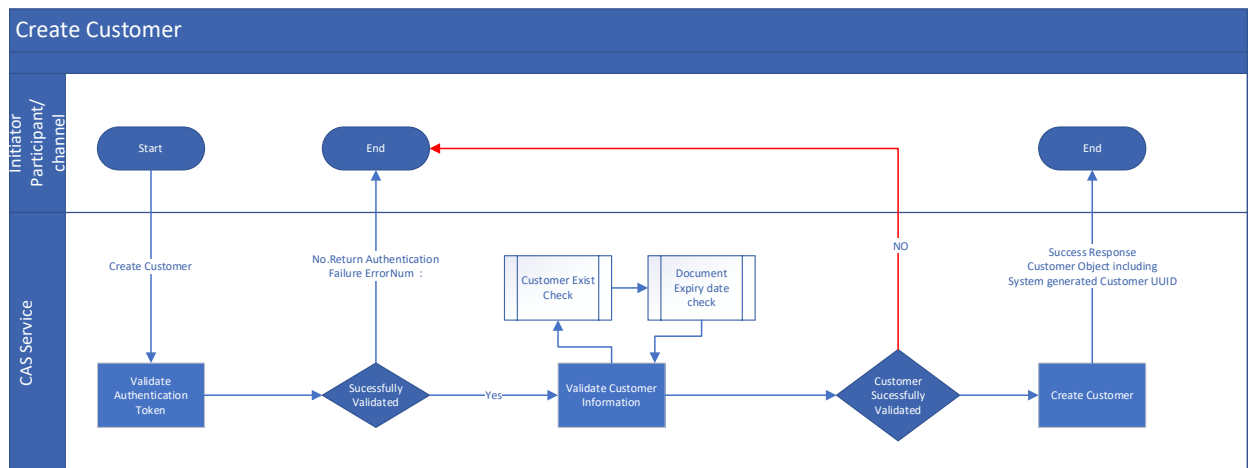


Figure 4-2 create customer design

4.2.1.1.2 API Definition

API Name	createCustomer
API Version	1.0.0
Endpoint	/cas/v2/customers
Method	POST

Table 5 create customer api definition

4.2.1.2 Update Customer

Use to update the customer entity. This will also create an archive record on the arcv_customer.

4.2.1.2.1 Use case scheme

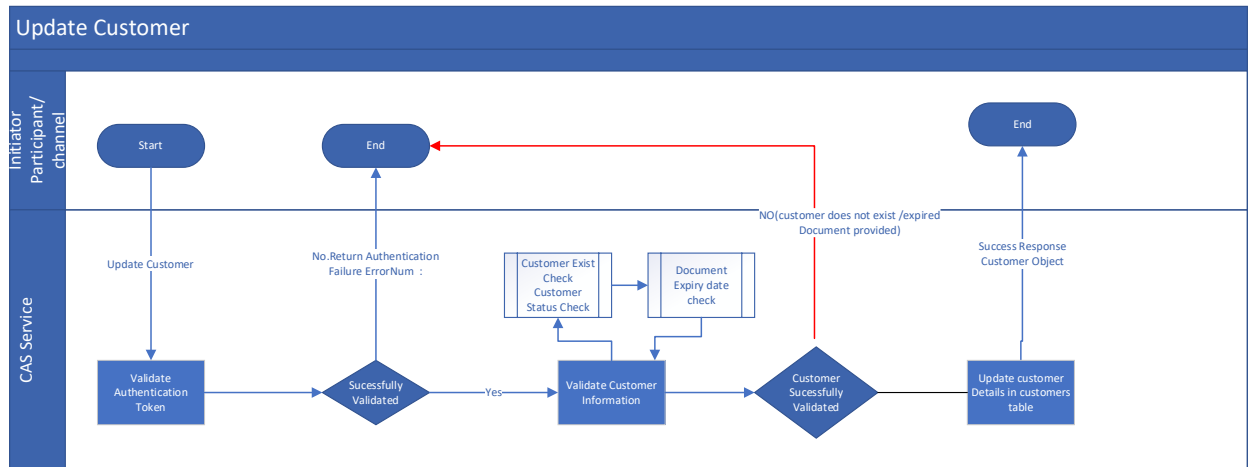


Figure 4-3 update customer design

4.2.1.2.2 API Definition

API Name	updateCustomer
API Version	1.0.0
Endpoint	/cas/v2/customers/{customer_uuid}
Method	PUT

Table 6 update customer api definition

4.2.1.3 Delete Customer

Use to delete the customer entity. All connected aliases and financial address are removed and become available for others to register anew.

4.2.1.3.1 Use case scheme

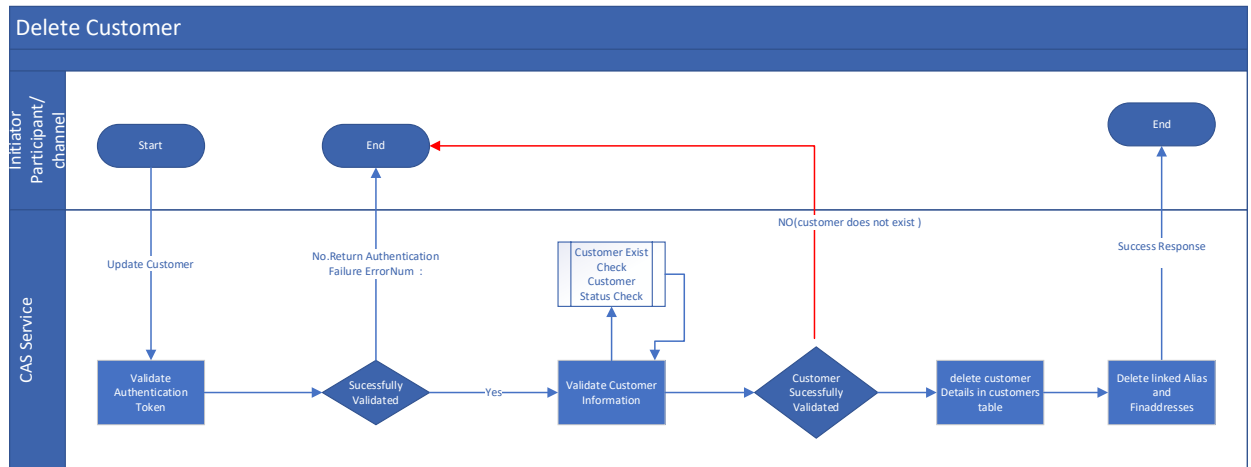


Figure 4-4 delete customer design

4.2.1.3.2 API Definition

API Name	deleteCustomer
API Version	1.0.0
Endpoint	/cas/v2/customers/{customer_uuid}
Method	DELETE

Table 7 delete customer api defintion

4.2.2 Alias Entity Management

4.2.2.1 Create Alias

This API is used to add an alias under the customer. When alias creation API is called `/api/fp/cas/v2/customers/{customer_uuid}/aliases`

SVIPS should first read the customer data (`GET/api/fp/cas/v2/customers/{customer_uuid}`)

we have scenarios according to customer type.

If customer type is equal to individual then IPS will create the Alias by following rule:

- @ as a prefix for a customer
- 10 first alpha characters composed of the beginning of the customer.lastName + 5 following characters for the customer.firstName, then 5 last characters corresponding to the last numbers of the identity document (CIN or passport)

For example: @RAMANITRERSERGE12345

If customer is company then IPS will register the Alias by following rule. Note that Alias value will be in free format.

- # as a prefix for a legal entity
- Alphanumeric characters evoking the customer.companyName

For example: #JIRAMAMGOFFICIAL

Note: For alias creation API `/api/fp/cas/v2/customers/{customer_uuid}/aliases`, `alias.type` and `alias.value` should be excluded/ignored from the incoming JSON

4.2.2.1.1 Use case scheme

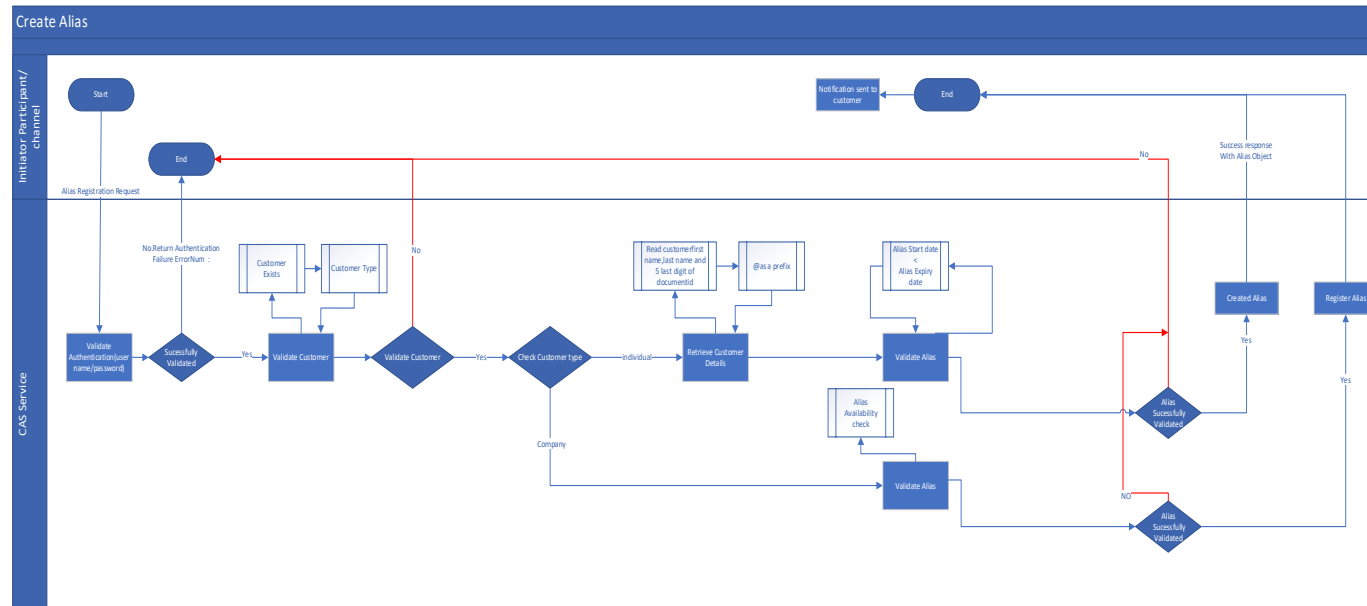


Figure 4-5 create alias design

4.2.2.1.2 API Definition

API Name	Create Alias
API Version	1.0.0
Endpoint	/cas/v2/customers/{customer_uuid}/aliases
Method	POST

Table 8 create alias api definition

4.2.2.2 Update Alias

This API is use to update Alias information. The Alias update is limited to field status , start_date ,expiry_date and is_owned.

4.2.2.2.1 Use case scheme

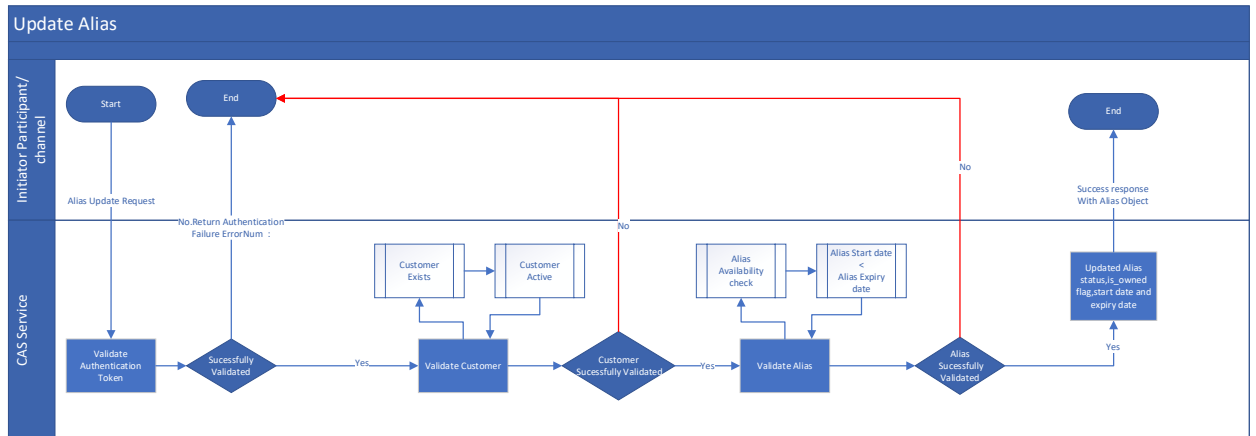


Figure 4-6 update alias design

4.2.2.2.2 API Definition

API Name	Update Alias
API Version	1.0.0
Endpoint	/cas/v2/customers/{customer_uuid}/aliases/{alias_uuid}
Method	PUT

Table 9 update alias api definition

4.2.2.3 Delete Alias

This API is use to remove an alias linked to the customer entity. Record pertaining to the Alias is deleted from defaultlinks.

Note: the finaddress associated with the alias is not deleted.

4.2.2.3.1 Use case scheme

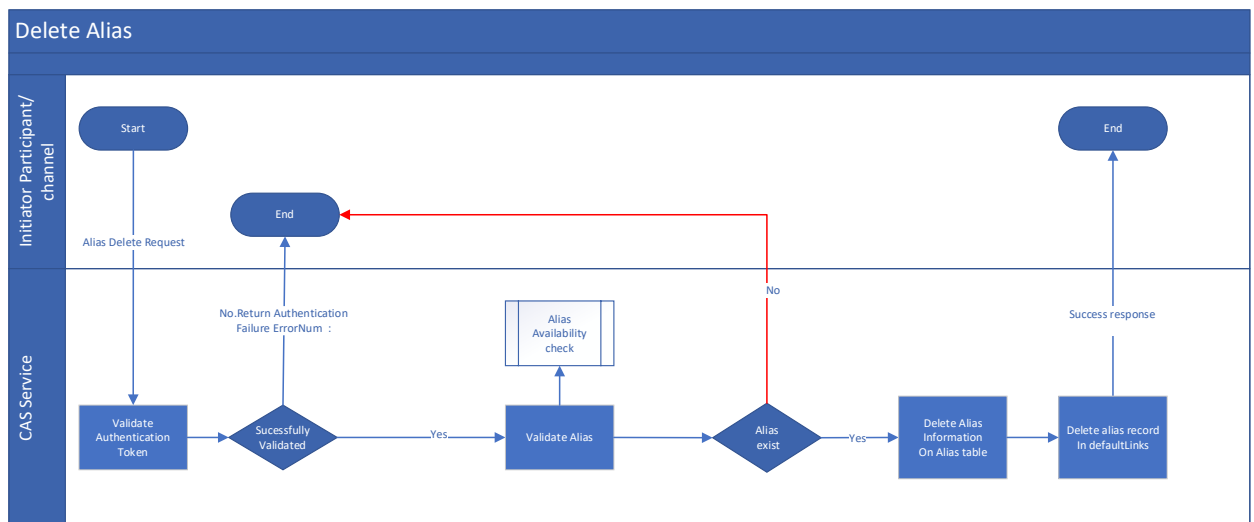


Figure 4-7 delete alias design

4.2.2.3.2 API Definition

API Name	Delete Alias
API Version	1.0.0
Endpoint	/cas/v2/customers/{customer_uuid}/aliases/{alias_uuid}
Method	DELETE

Table 10 delete alias api definition

4.2.3 FinAddress Entity Management

4.2.3.1 Create Finaddress

This Api is use to add a finAddress under a customer.

4.2.3.1.1 Use case scheme

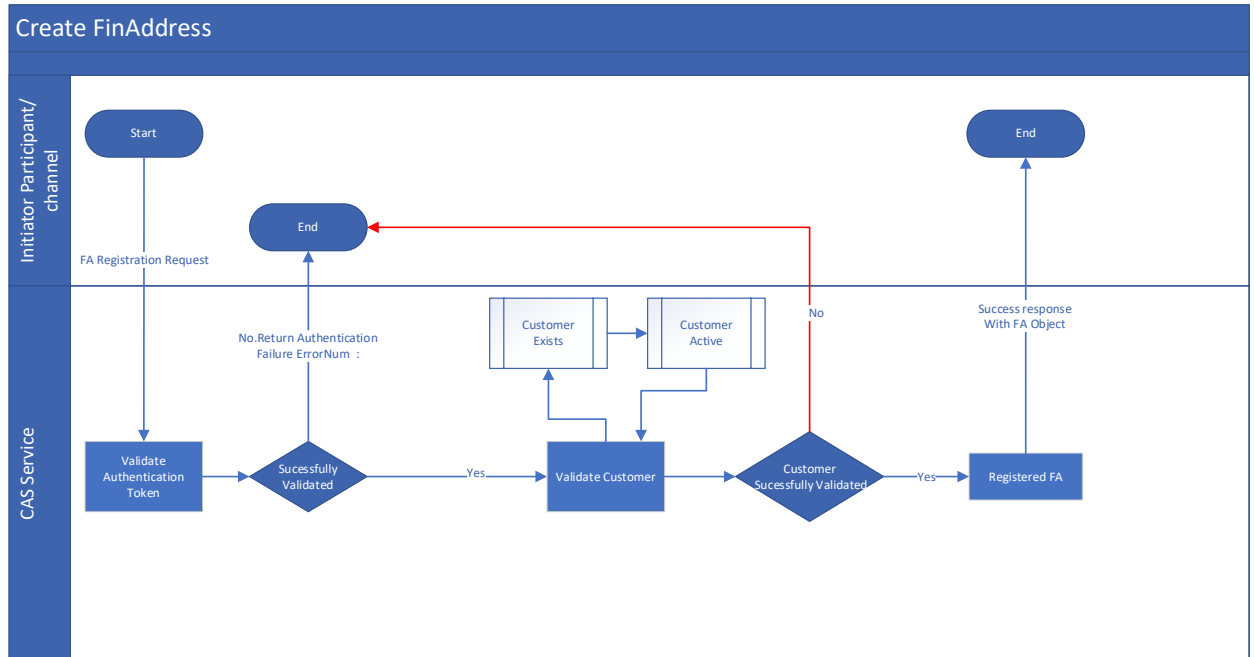


Figure 4-8 create finaddress design

4.2.3.1.2 API Definition

API Name	Create Finaddress
API Version	1.0.0
Endpoint	/cas/v2/customers/{customer_uuid}/finaddresses
Method	POST

Table 11 create finaddress api definition

4.2.3.2 Delete FinAddress

This Api is use to delete a finAddress under a customer.

4.2.3.2.1 Use case scheme

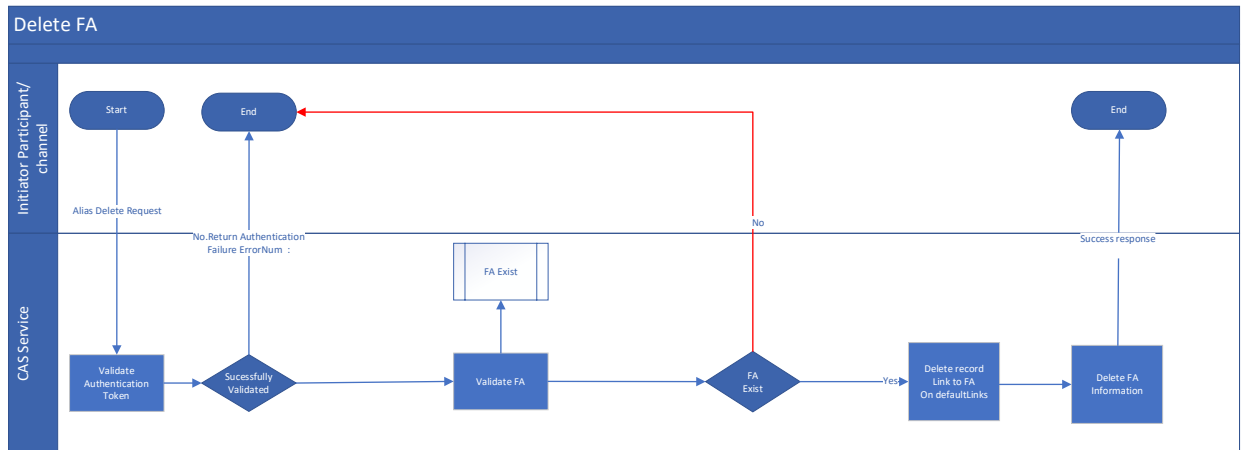


Figure 4-9 delete finaddress design

4.2.3.2.2 API Definition

API Name	Delete Finaddress
API Version	1.0.0
Endpoint	/cas/v2/customers/{customer_uuid}/finaddresses/{finaddress_uuid}
Method	POST

Table 12 delete finaddress api definition

4.2.3.3 Set FinAddress to default

This API is use to set the finaddress set as the default finaddress for the customer.

4.2.3.3.1 For Customer with lineage_type is Individual

Use case scheme

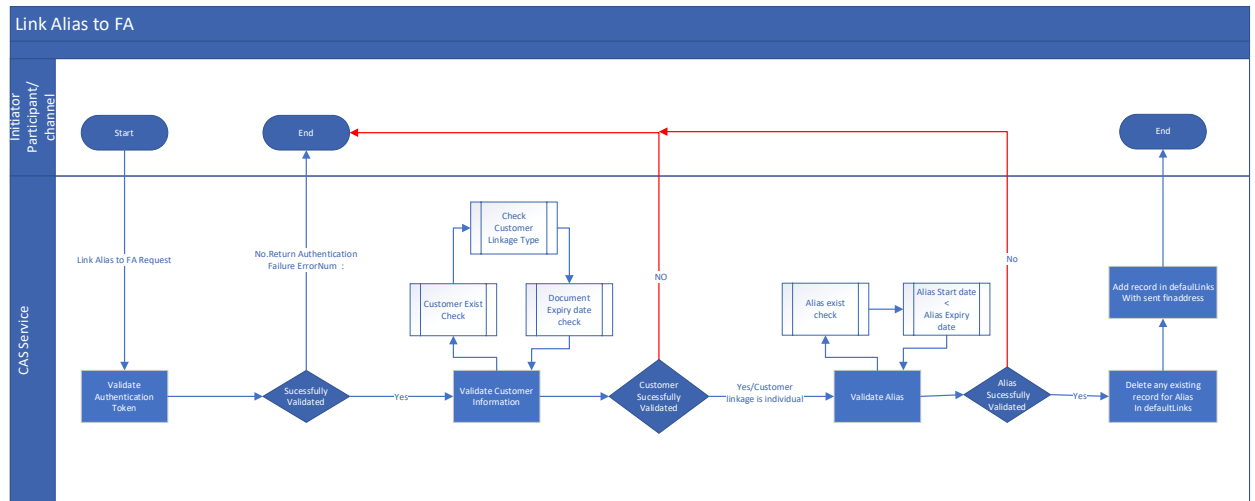


Figure 4-10 Customer with lineage_type is Individual design

API Definition

API Name	Link Alias to FA(linkage type=individual)
API Version	1.0.0
Endpoint	/cas/v2/customers/{customer_uuid}/aliases/{alias_uuid}/finaddresses/{finaddress_uuid}/default
Method	POST

Table 13 Customer with lineage_type is Individual api defintion

4.2.3.3.2 For Customer with Linkeage_type is Universal

Use case scheme

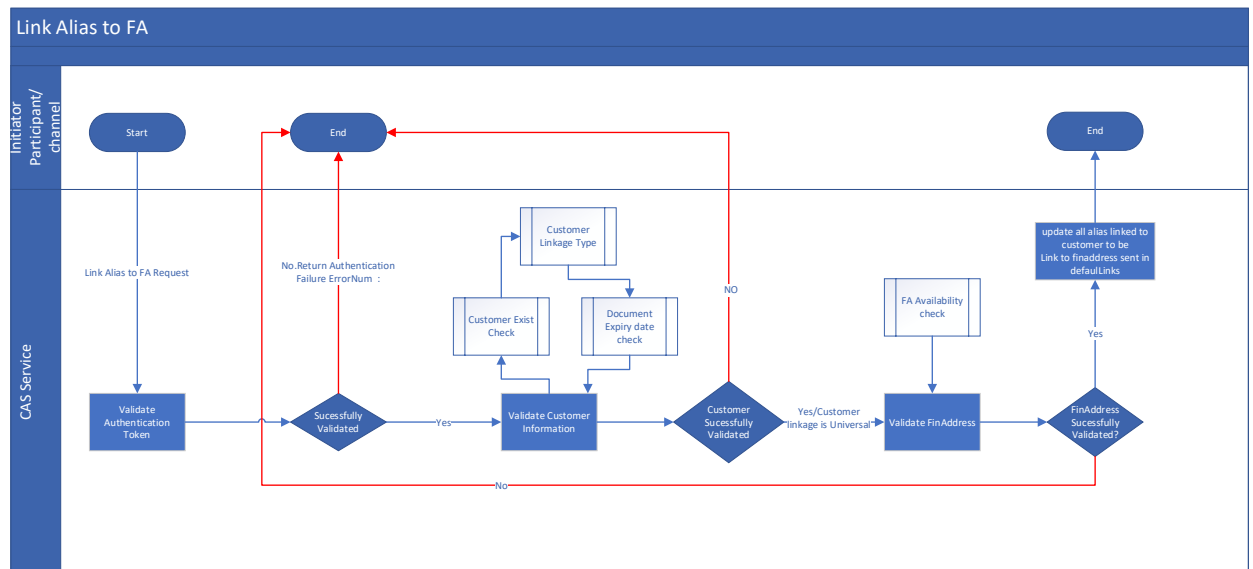


Figure 4-11 Customer with linkeage_type is universal design

API Definition

API Name	Link Alias to FA(linkage type= universal)
API Version	1.0.0
Endpoint	/cas/v2/customers/{customer_uuid}/finaddresses/{finaddress_uuid}/default
Method	POST

Table 14 Customer with linkeage_type is universal api definition

4.3 Alias Resolution

Alias resolution is use by a participant to resolve an alias to the default fin address before initiating an RTP or Payment message (In the case of request to pay /payment by Alias)

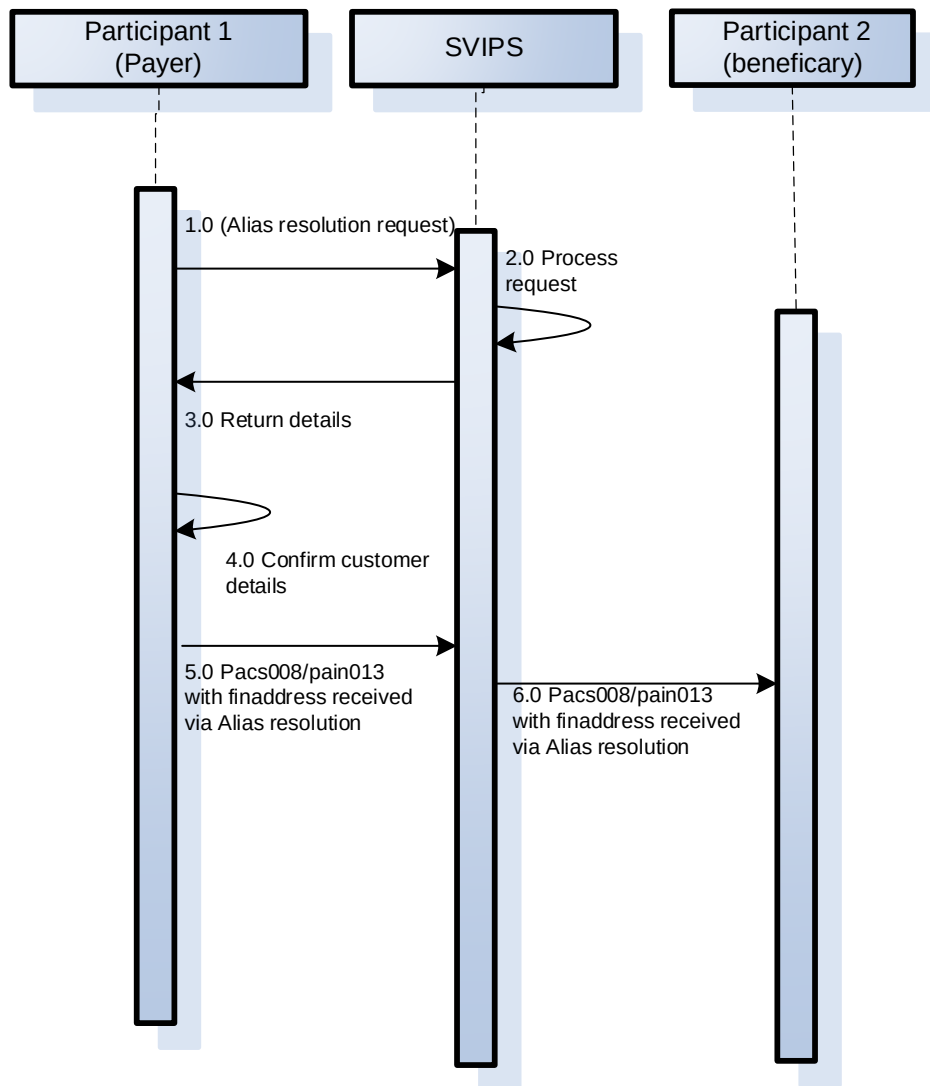


Figure 4-12 alias resolution with payment

4.3.1 Use case scheme

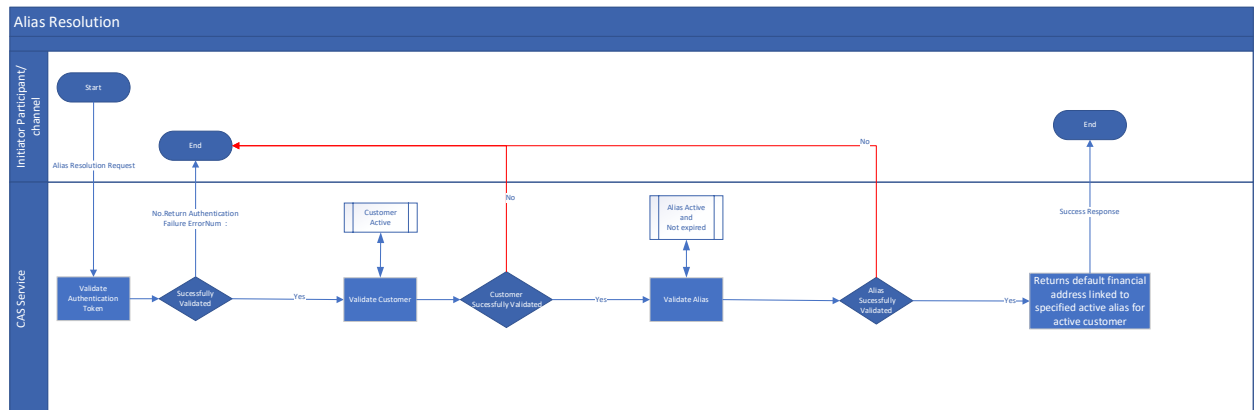


Figure 4-13 alias resolution design

4.3.2 API Definition

API Name	Alias Resolution
API Version	1.0.0
Endpoint	/cas/v2/finaddresses?alias_type=&alias_value=
Method	GET

Table 15 alias resolution api resolution

4.4 Customer Creation with OTP

4.4.1 OTP generation Step

First an OTP generation request is sent to SVIPS. SVIPS generate a random 6 digits number and set the validityDatetime. The OTP is linked to the customer uuid send in the request.

Each call will generate a new OTP and invalidate previous OTP of the specific customer. Every OTP code will have a 15 minutes lifetime after which is becomes automatically invalidated

Same OTP API can be used by participant to validate any user step. e.g. details update, creating/updating alias, etc.

4.4.1.1 Use case scheme

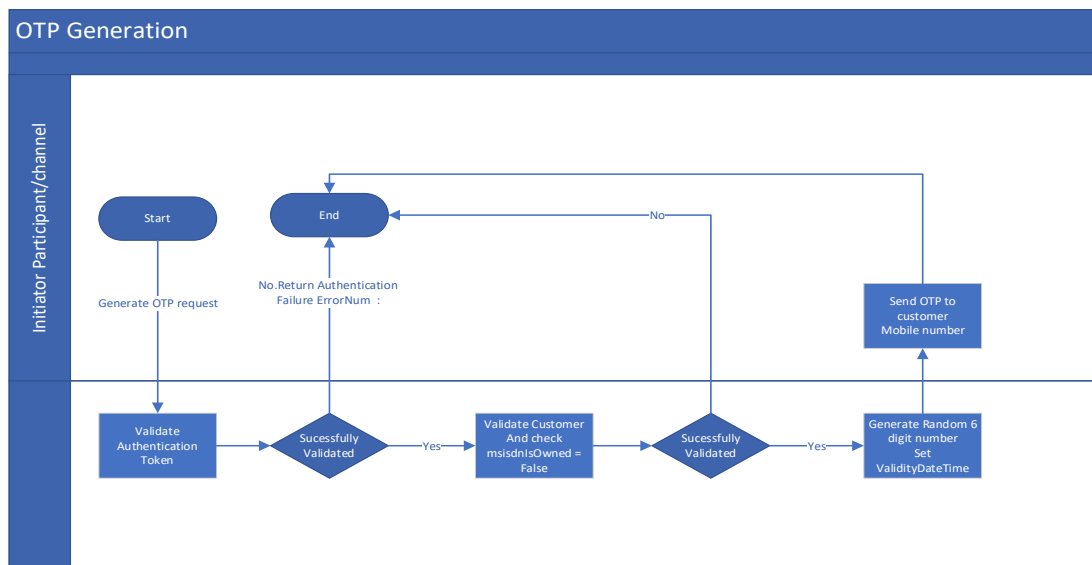


Figure 4-14 otp generation design

4.4.1.2 API Definition

API Name	OTP generation
API Version	1.0.0
Endpoint	cas/v2/customers/{customer_uuid}/msisdn_otp
Method	GET

Table 16 otp generation api definition

4.4.2 Update customer Status and set msisdnsOwned =true

Once the customer receives the OTP, in his mobile, he will enter the OTP on participant side and participant will sent the update request with the OTP in the payload.

4.4.2.1 Use case scheme

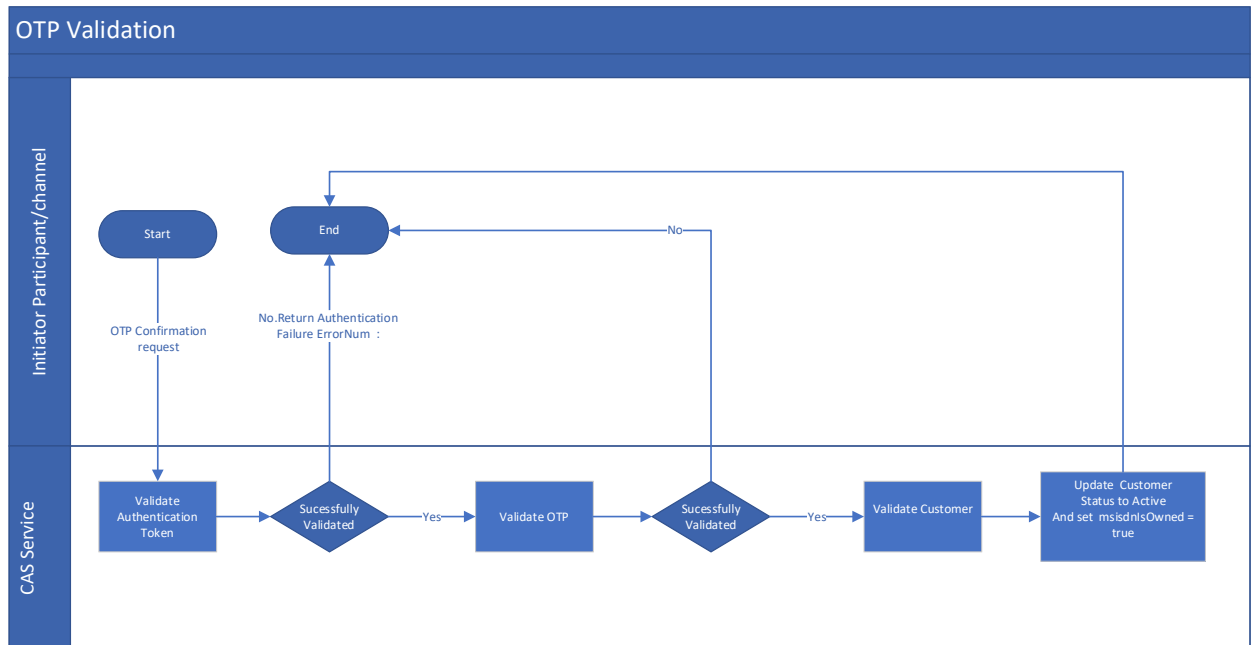


Figure 4-15 otp validation design

Note: One OTP of specific customer can be attempted 3 times, if exceed 3 times, it will be automatically invalidated

4.4.2.2 API Definition

API Name	OTP validation
API Version	1.0.0
Endpoint	cas/v2/customers/{customer_uuid}/msisdns_otp
Method	POST

Table 17 otp validation api definition

5 API Reference

The CAS APIs detailed information provided in the attached [JSON file](#) in reference table, which can be edited using any open APIs editor.

There are many editors online and offline which are free to use like swagger, postman, redocly etc.

Let's take example of swagger editor:

Access the Editor: Go to Swagger Editor from browser(<https://editor-next.swagger.io/>).

Load or Create API Definition: Open an OpenAPI definition file provided in reference table.

Preview: View the live preview of the API documentation on the right side. As shown below.

POST /cas/v2/customers/one-step

Parameters

Try it out

No parameters

Request body required

application/json

Example Value | Schema

```
{
  "customer": {
    "firstName": "string",
    "lastName": "string",
    "gender": "MALE",
    "dob": "2024-06-10T12:15:45.113Z",
    "externalId": "string",
    "linkageType": "INDIVIDUAL_DEFAULT",
    "documentType": "NATIONAL_ID",
    "documentId": "string",
    "documentValidityDate": "2024-06-10T12:15:45.113Z",
    "msisdn": "+167-+6+8(18"
  },
  "alias": {
    "type": "MSISDN",
    "value": "string",
    "startDate": "2024-06-10T12:15:45.113Z",
    "expireDate": "2024-06-10T12:15:45.113Z"
  },
  "finAddress": {
    "type": "IBAN",
    "finAddress": "string",
    "servicerIdType": "string",
    "servicerId": "string"
  }
}
```

Activate Windows
Go to Settings to activate Windows.

Responses

Code	Description	Links
200	OK Media type application/json Controls Accept header. Example Value Schema <pre>{ "message": "string", "responseCode": 0, "errorMessage": "string", "errorDescription": "string", "success": true, "response": {} }</pre>	No links
400	Reference to Enum schema Media type application/json Example Value Schema <pre>"ERR001"</pre>	No links
500	Internal Server Error Activate Windows Go to Settings to activate Windows.	No links

5.1 API LIST

5.1.1 Customers management

Register a new customer:

POST /cas/v2/customers

Register a new customer, 1 alias and 1 default financial address in one step:

POST /cas/v2/customers/one-step

get customer data:

GET /cas/v2/customers/{customer_uuid}

Update customer data and status:

PUT /cas/v2/customers/{customer_uuid}

Deregister customer:

DELETE /cas/v2/customers/{customer_uuid}

5.1.2 Aliases management

Add a new alias:

POST /cas/v2/customers/{customer_uuid}/aliases

get alias data:

GET /cas/v2/customers/{customer_uuid}/aliases/{alias_uuid}

Update alias data and status:

PUT /cas/v2/customers/{customer_uuid}/aliases/{alias_uuid}

Remove alias:

DELETE /cas/v2/customers/{customer_uuid}/aliases/{alias_uuid}

5.1.3 Financial addresses management

Add a new financial address:

POST /cas/v2/customers/{customer_uuid}/finaddresses

get financial address data:

GET /cas/v2/customers/{customer_uuid}/finaddresses/{finaddress_uuid}

Make financial address a default address:

customer.linkage_type=Universal_default

POST
/cas/v2/customers/{customer_uuid}/finaddresses/{finaddress_uuid}/default

customer.linkage_type=Individual_default

POST
/cas/v2/customers/{customer_uuid}/aliases/{alias_uuid}/finaddresses/{finaddress_uuid}/default

Remove financial address:

DELETE /cas/v2/customers/{customer_uuid}/finaddresses/{finaddress_uuid}

Resolve default alias:

GET /cas/v2/finaddresses?alias_type=&alias_value=

6 Reference

Reference	File
API Authorization chapter 3	SVIPS_Specifications_sv_ip__how-to_specification_en_core
API reference chapter 5	 BPC FP service APIs.json